



FluxLink

VE441 · App Development for Entrepreneurs · Project Proposal

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UM-SJTU Joint Institute · Shanghai Jiao Tong University

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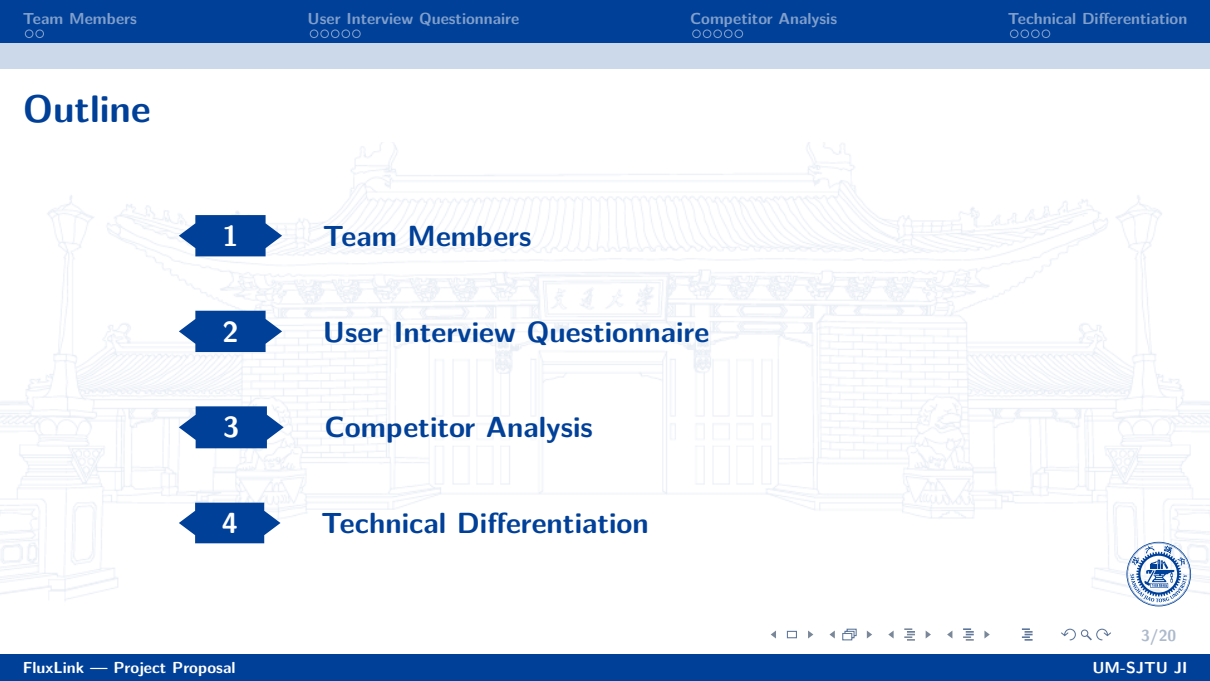
FluxLink — Elevator Pitch

*Never waste time choosing how to send a file again —
FluxLink automatically selects the fastest, most secure transfer path
across any platform and any network.*

- **Zero decision cost** — routing is automatic
- **Any network** — LAN, P2P, relay fallback
- **Any platform** — Android, iOS, Desktop
- **Uniform E2EE** — all modes, all files
- **Encrypted chat** — context stays private
- **No account required** — pair via QR/token



Outline

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- 1 Team Members
 - 2 User Interview Questionnaire
 - 3 Competitor Analysis
 - 4 Technical Differentiation



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Technical Differentiation



Team Members

Team Name: NUUB

PM: Zhang Jinbin

Full Name	jAccount	Task Assignment	Technical Strengths
Sun Qizhen	qz_sun	Product architecture · Protocol routing engine · Network path scoring	Networking · Distributed systems · System design
Zhang Jinbin	moranxuege	Mobile client (Android/iOS) · Cross-platform UI/UX	Android (Kotlin) · iOS (Swift) · UX implementation
Li Yuhao	y4954he	End-to-end encryption · QR/token handshake · Encrypted chat	Cryptography · Backend security · Protocol design
He Yumeng	17521218310	Relay server · Deployment · Testing · Cache policy	Backend · DevOps · QA · Django REST

Product & Research

Sun Qizhen
routing logic &
protocol priority

Client & UX

Zhang Jinbin
mobile UI &
cross-platform

Networking & Security

Li Yuhao
E2EE model &
handshake

Backend & QA

He Yumeng
relay infra & test
coverage



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Interview Design — Participants & Venue

How We Find Interviewees

- SJTU engineering/CS communities
- GitHub discussions, V2EX, WeChat tech groups
- Screening: regularly transfer files across multiple devices/OS, and have encountered at least one of: size limits, incompatibility, network failure, privacy concerns
- **Avoid friends/family** — reduce social desirability bias

Venue & Session Structure

- Remote via Tencent Meeting / Zoom
 - 10–20 min per session
 - One interviewer + one note-taker
 - Recorded with consent
- | | |
|-----------|---------------------------------|
| 0–5 min | Warm-up, device context |
| 5–15 min | Pain stories, failure scenarios |
| 15–20 min | Ideal workflow, retention |



Theme 1: Device Context & Current Behavior

Goal: Establish devices/platforms used and current tools, without leading toward any product concept.

- Q1** Walk me through the devices you use daily and how often they need to share files with each other.
- Q2** Think back to the last few weeks — what triggered your need to send a file, and to whom?
- Q3** What apps or methods do you normally reach for? How did you settle on those tools?
- Q4** When you're about to send a file, what goes through your head before deciding which tool to use?



Theme 2: Pain Points & Failure Stories

Goal: Surface real breakdown moments, workarounds, and emotional friction.

- Q5** Walk me through the last time you sent a large file — from the moment you decided to send it to delivery.
- Q6** Has your usual method ever just stopped working? What happened, and what did you do instead?
- Q7** When transferring between iPhone and Android, or phone and a different-OS computer, how do you handle that?
- Q8** Describe a time when security/privacy crossed your mind during a file transfer. Did it change your behavior?
- Q9** What does it feel like when a transfer fails or arrives corrupted? Tell me about a specific moment.



Theme 3: Motivations, Ideals & Accepted Friction

Goal: Understand retention drivers, unarticulated needs, and normalized workarounds.

- Q10** When a file transfer tool works really well, what makes you keep coming back to it?
- Q11** Describe your ideal file-sending experience for a scenario that comes up regularly for you.
- Q12** What frustrations have you accepted as “the way file transfer works” — things you’ve stopped trying to fix?



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Market Overview

The Problem with Existing Tools

The file transfer market appears saturated, but **no mainstream product** solves the combined problem of:

- Cross-platform compatibility
- Adaptive network routing
- Consistent end-to-end encryption
- Low user decision cost

in a single experience.

Competitors analyzed: WeTransfer · Snapdrop · SHAREit · Send Anywhere · Xender · LocalSend (emerging reference)



Competitor Profiles (1/2)

WeTransfer — Web-first Cloud Sharing

- + Simple UI; no recipient install; async delivery; 50M+ downloads; 4.2🤖
- Internet required; no P2P/LAN; 2 GB free limit; no E2EE; no auto-routing

Snapdrop — Browser-based LAN Sharing

- + Zero install; cross-platform via browser; no account; open source; 4.0🤖
- LAN-only; browser fragility; no fallback; no transfer recovery

SHAREit — Mobile-first High-speed Local

- + Extremely fast; 1B+ downloads; cross-platform; handles large files; 3.8🤖
- Ad-heavy; documented privacy/security issues; no E2EE; trust erosion



Competitor Profiles (2/2)

Send Anywhere — Key-based Transfer

- + 6-digit key model; wide platform support; encrypted positioning; 4.3 🤖
- No auto-routing; relay-dependent in real networks; no encrypted chat

Xender — Group Sharing

- + Fast WiFi Direct; 500M+ downloads; group sharing; 4.0 🤖
- Ad-heavy; no E2EE; limited to local network; no relay fallback

LocalSend — Open-source LAN Utility

- + Clean; privacy-friendly; no account; open source (MIT); 4.7 🤖
- Strictly LAN-bound; no NAT traversal; no relay fallback



Comparison Table

Feature	WeTransfer	Snapdrop	SHAREit	Send Anywhere	Xender	LocalSend	FluxLink
Cross-platform	✓	✓	✓	✓	✓	✓	✓
LAN transfer	×	✓	✓	Partial	✓	✓	✓
P2P (no server)	×	Partial	✓	Partial	✓	✓	✓
Internet/relay fallback	✓	×	×	✓	×	×	✓
Auto-routing	×	×	×	×	×	×	✓
E2EE (all modes)	×	Partial	×	Partial	×	✓ (LAN)	✓
Encrypted chat	×	×	×	×	×	×	✓
No account required	✓	✓	×	✓	×	✓	✓
Ad-free	✓	✓	×	Partial	×	✓	✓
Google Play rating	4.2	4.0	3.8	4.3	4.0	4.7	—



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Intelligent Transfer Routing Engine

Core Insight

Every competitor optimizes a **single transfer modality** and asks users to manage the rest manually. FluxLink treats the protocol stack as a **scored decision**, not a user configuration.

Priority chain (probed automatically after pairing):

- ① **mDNS-based LAN** — lowest latency, highest throughput, no internet dependency
- ② **Magic Wormhole P2P** — rendezvous-server-assisted NAT traversal; no file content on server
- ③ **IPv6 direct P2P** — increasingly viable on campus and carrier networks
- ④ **Centralized relay fallback** — guarantees delivery under any condition; relay handles ciphertext only

*The user sees one consistent interaction: **pair, send, done**. Path degradation triggers silent fallback.*



Security Model & Encrypted Chat

Uniform Security Across All Modes

- QR / token handshake establishes a session key
- All file chunks and chat messages encrypted **before leaving the device**
- Relay server never sees plaintext
- Relay-cached messages cleared weekly

Encrypted Chat Layer

- **No competitor** integrates encrypted chat alongside file transfer
- The conversation about a file is as private as the file itself
- Same E2EE session — no context leakage

Why This is Hard to Copy

The routing engine cannot be added incrementally to a single-modality app. Competitors would need to simultaneously rebuild their network layer, session model, and security architecture — at odds with their current business models and architectural philosophy.



Summary

What We Bring

- Intelligent multi-protocol routing engine
- Uniform E2EE across LAN, P2P, and relay
- Integrated encrypted chat
- No account, no ads, no decision cost

Next Steps

- Conduct user interviews (10–15 participants)
- Synthesize pain points & refine problem framing
- Prototype routing engine core
- Validate pairing UX with paper prototypes

Thank you — Questions?





Thank You